Diabetic Retinopathy Digital Twin

AI-Powered Disease Progression Forecasting & Personalized Screening

Digital Twin Challenge 2026 — Happiest Health Submission

Domain: Healthcare AI, Vision Deep Learning & Physiological Simulation

Team: Digital Twin Innovator | Public Open-Source Submission

Problem Statement & Rural Healthcare Crisis

The Preventable Blindness Epidemic Across Rural India

- **India's Diabetic Burden**
  80+ million diabetic adults — 2nd highest globally. ~18% develop Diabetic Retinopathy (DR).
- **Preventable Vision Loss**
  Early regular screening prevents up to 90% of severe vision loss and blindness.
- **Severe Specialist Shortage**
  Barely 1 ophthalmologist per 100,000 rural population. Manual mass screening is mathematically impossible.
- **Deficiencies of Existing AI**
  Existing algorithms operate as one-shot black boxes, fail on noisy portable fundus cameras, and lack longitudinal progression modeling.

Healthcare Use Case & Clinical Value Proposition

Transforming Reactive Eye Checks into Proactive Digital Twin Intelligence

- **Patient-Centric Virtual Replica**
  Synthesizes fundus imaging, historical EHR labs, and continuous glucose wearable streams into a dynamic twin.
- **Primary Health Centre (PHC) Empowerment**
  Enables rural health workers to capture fundus photos with handheld cameras and receive instant clinical triage.
- **Dynamic Progression Trajectories**
  Forecasts how retinopathy severity will evolve over 1-5 years using Monte Carlo Markov models.
- **Personalized Screening Optimization**
  Transitions from rigid 1-year schedules to risk-adaptive intervals (24 mo down to immediate urgent referral), maximizing specialist bandwidth.

End-to-End System Architecture

Modular Multi-Modal Pipeline: Ingestion, Vision AI, Twin Simulation, and Interactive UI

- **1. Multi-Modal Ingestion**
  Fundus photographs + Longitudinal EHR (HbA1c, BP, eGFR, meds) + 5-minute CGM wearable streams.
- **2. Robust Vision Preprocessing**
  Circular ROI extraction, Ben Graham color normalization, and LAB CLAHE contrast enhancement for field cameras.
- **3. Deep Learning Vision Core**
  Pretrained EfficientNet-B3 backbone with custom ordinal head + GradCAM visual explanation hooks.
- **4. Digital Twin Engine**
  Multi-factor clinical risk engine (0-100) + 5-state Markov stochastic progression model.
- **5. Interactive Dashboard**
  Full Streamlit clinician dashboard with interactive Plotly risk gauge, radar charts, CGM curves, and forecast areas.

AI/ML Model & Explainability Framework

Deep Transfer Learning Paired with Trustworthy Explainable AI (XAI)

- **Vision Classifier Architecture**

  EfficientNet-B3 backbone pretrained on ImageNet, custom linear-ReLU-dropout classification head.

- **Ordinal ICDR Scale**

  Evaluates 5 standardized clinical grades: Grade 0 (No DR) to Grade 4 (Proliferative DR).

- **Field Camera Robustness**

  Trained with Albumentations: Gaussian blurs, coarse dropouts (simulating dust), rotations, and color shifts.

- **Explainable AI (GradCAM)**

  Generates gradient-weighted activation maps to highlight microaneurysms, hemorrhages, and exudates directly on the fundus photograph.

Digital Twin Engine & Progression Modeling

Evidence-Weighted Risk Stratification & Longitudinal Markov Simulation

- **8 Clinical Risk Factors**

  Weights grounded in UKPDS, DCCT, and WESDR landmark trials (DR Grade 30%, HbA1c 20%, Duration 15%, BP 10%, CGM TIR 10%, etc.

- **Markov Monte Carlo Simulation**

  Simulates 1,000 stochastic 5-year patient trajectories using transition matrices dynamically adjusted by patient risk multiplier.

- **Wearable CGM Telemetry**

  Models circadian glucose variability, Dawn Phenomenon, and Time-in-Range (70-180 mg/dL target) from wearable data.

- **Adaptive Screening Engine**

  Assigns personalized screening intervals (24, 12, 6, 3 months, or Emergency Referral) and tailored lifestyle guidance.

Technical Stack & Verification Summary

Modern Open-Source Technologies Built for Portability and Speed

- **Core Programming & Frameworks**
  Python 3.10+, PyTorch 2.0+, TIMM, Albumentations, OpenCV, NumPy, SciPy, Pandas.
- **Clinical Dashboard UI**
  Streamlit with custom dark medical CSS + Plotly interactive SVG rendering.
- **Benchmarking Datasets**
  APTOS 2019 Blindness Detection, IDRiD (Indian diabetic cohort), DRIVE (vessels), Messidor-2.
- **Verification Suite**
  Automated test suite (tests/test_core.py) validating end-to-end data pipelines, risk scoring, progression, and twin payload packaging.

Project Outcomes & Rural Deployment Path

Immediate Hackathon Deliverable and Future Health System Integration

- **Ready-to-Use Artifacts**

  Complete open-source GitHub repository, running interactive dashboard, sample fundus images, and Kaggle training notebook.

- **Tele-Ophthalmology Synergy**

  Designed for deployment at Ayushman Bharat Health and Wellness Centres (AB-HWCs) across rural India.

- **ABDM & HL7 FHIR Interoperability**

  Data schemas aligned with Ayushman Bharat Digital Mission (ABDM) standards for frictionless EHR linkage.

- **Ethical & Safe AI**

  Incorporates XAI visual evidence, confidence thresholds, and mandatory specialist referral overrides for advanced stages.